

STANDARD 2

Stage 6

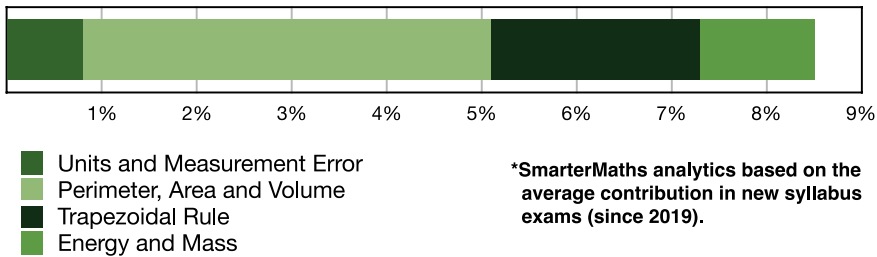
Measurement (Std 2), M1 Applications of Measurement (Y11)

Perimeter, Area and Volume (Std 2)

Teacher: M K

Exam Equivalent Time: 145.5 minutes (based on allocation of 1.5 minutes per mark)

M1 Applications of Measurement



IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- M1 Applications of Measurement* has contributed a healthy average of 8.5% per new syllabus Std2 exam since it was introduced in 2019.
- We have split this area into 4 categories for the purpose of analysis: 1-Units and Measurement Error (0.8%), 2-Perimeter, Area and Volume (4.3%), 3-Trapezoidal Rule (2.2%) and 4-Energy and Mass (1.2%).
- This analysis looks at *Perimeter, Area and Volume*.

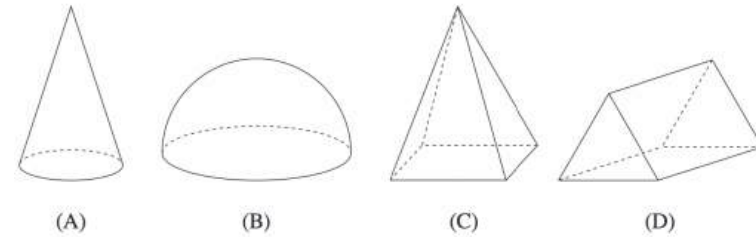
ANALYSIS - Common pitfalls

- Perimeter, Area and Volume* has contributed 5 separate longer answer questions in the 2022-24 period, regularly producing sub-50% mean marks. This is after the 2021-2019 Std2 papers presented low difficulty examinations of the topic area.
- This topic area is very broad and to revise effectively, there is no substitute to doing lots of varied past exam questions.
- Well designed revision must include a focus on area and volume questions that incorporate circular measure which are easily the most commonly asked and poorly answered Most recently 2024 Std2 34 illustrates this key revision example.
- Water catchment questions (examined in 2022 for the first time since 2016) have caused problems and deserve attention.

Questions

1. Measurement, STD2 M1 2011 HSC 1 MC

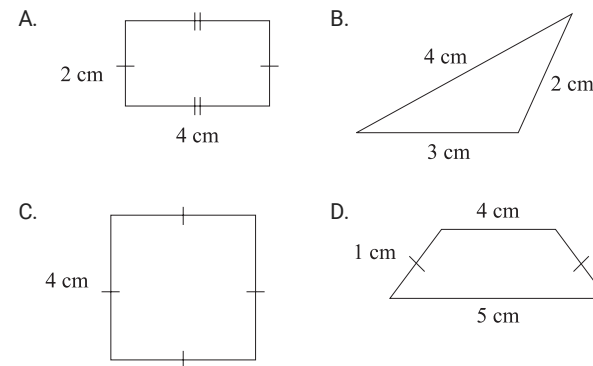
Which of the solids shown is a prism?



2. Measurement, STD2 M1 2019 HSC 1 MC

⚡ RAP Data (vs state mean) - Bottom 19%: School result (98%) was 4% above state average (94%)

Which of the following shapes has a perimeter of 12 cm?

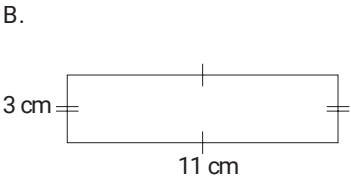
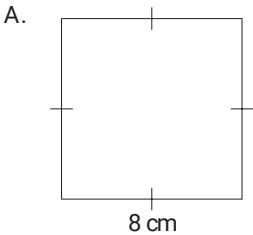


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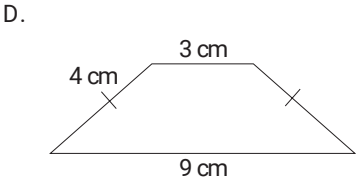
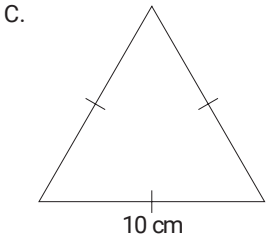
3. Measurement, STD2 M1 2021 HSC 1 MC

⚡ *RAP Data (vs state mean) - Bottom 16%: School result (94%) was 3% above state average (91%)*

Which of the following shapes has the largest perimeter?

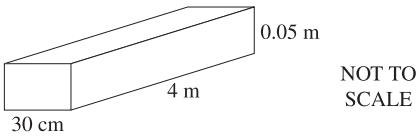


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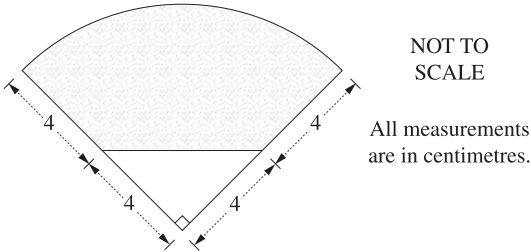
4. Measurement, STD2 M1 2012 HSC 6 MC

What is the volume of this rectangular prism in cubic centimetres?



- A. 6 cm³
- B. 600 cm³
- C. 60 000 cm³
- D. 6 000 000 cm³

5. Measurement, STD2 M1 2009 HSC 11 MC

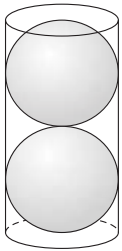


What is the area of the shaded part of this quadrant, to the nearest square centimetre?

- A. 34 cm²
- B. 42 cm²
- C. 50 cm²
- D. 193 cm²

6. Measurement, STD2 M1 2009 HSC 19 MC

Two identical spheres fit exactly inside a cylindrical container, as shown.



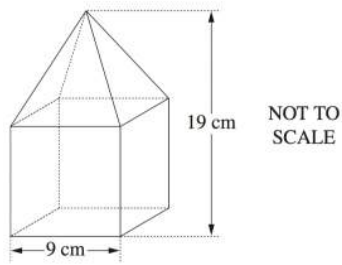
The diameter of each sphere is 12 cm.

What is the volume of the cylindrical container, to the nearest cubic centimetre?

- A. 1357 cm³
- B. 2714 cm³
- C. 5429 cm³
- D. 10 857 cm³

7. Measurement, STD2 M1 2013 HSC 12 MC

A square pyramid fits exactly on top of a cube to form a solid.

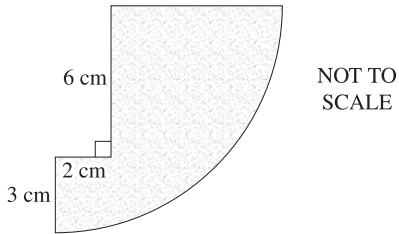


What is the volume of the solid?

- A. 513 cm³
- B. 999 cm³
- C. 1242 cm³
- D. 1539 cm³

8. Measurement, STD2 M1 2013 HSC 16 MC

The shaded region shows a quadrant with a rectangle removed.

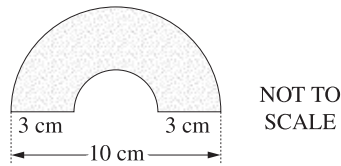


What is the area of the shaded region, to the nearest cm²?

- A. 38 cm²
- B. 52 cm²
- C. 61 cm²
- D. 70 cm²

9. Measurement, STD2 M1 2013 HSC 19 MC

A logo is designed using half of an annulus.



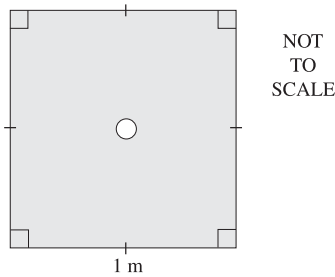
What is the area of the logo, to the nearest cm²?

- A. 25 cm²
- B. 33 cm²
- C. 132 cm²
- D. 143 cm²

10. Measurement, STD2 M1 2008 HSC 11 MC

The diagram shows the floor of a shower. The drain in the floor is a circle with a diameter of 10 cm.

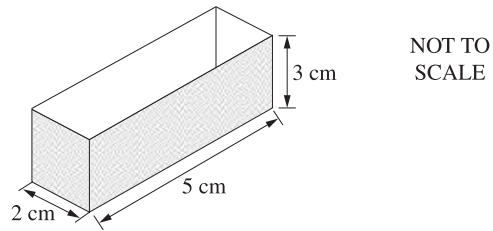
What is the area of the shower floor, excluding the drain?



- A. 9686 cm²
- B. 9921 cm²
- C. 9969 cm²
- D. 10 000 cm²

11. Measurement, STD2 M1 2008 HSC 2 MC

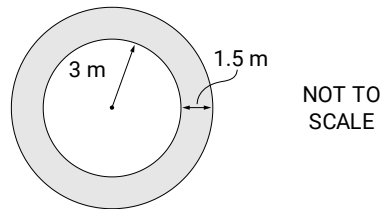
What is the surface area of the open box?



- A. 10 cm^2
- B. 30 cm^2
- C. 52 cm^2
- D. 62 cm^2

12. Measurement, STD2 M1 2014 HSC 12 MC

A path 1.5 metres wide surrounds a circular lawn of radius 3 metres.

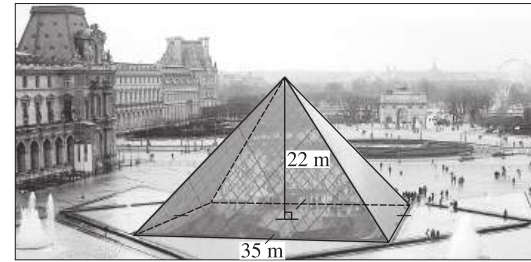


What is the approximate area of the path?

- A. 7.1 m^2
- B. 21.2 m^2
- C. 35.3 m^2
- D. 56.5 m^2

13. Measurement, STD2 M1 2015 HSC 8 MC

The Louvre Pyramid in Paris has a square base with side length 35 m and a perpendicular height of 22 m.

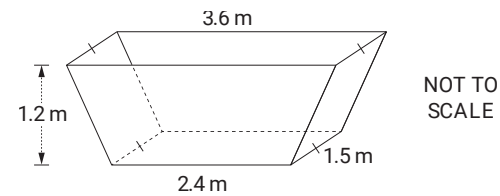


What is the volume of this pyramid, to the nearest m^3 ?

- A. 257 m^3
- B. 1027 m^3
- C. 8983 m^3
- D. $26\,950 \text{ m}^3$

14. Measurement, STD2 M1 2017 HSC 18 MC

A skip bin is in the shape of a trapezoidal prism, with dimensions as shown.

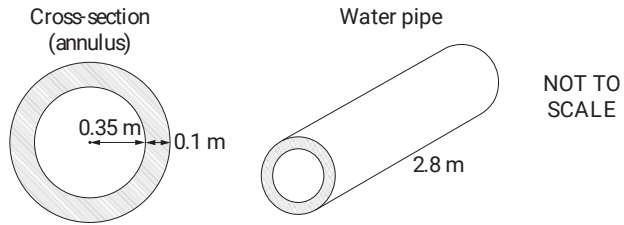


What is the volume of the skip bin?

- A. 5.4 m^3
- B. 7.776 m^3
- C. 10.8 m^3
- D. 15.552 m^3

15. Measurement, STD2 M1 2017 HSC 22 MC

A concrete water pipe is manufactured in the shape of an annular cylinder. The dimensions are shown in the diagrams.

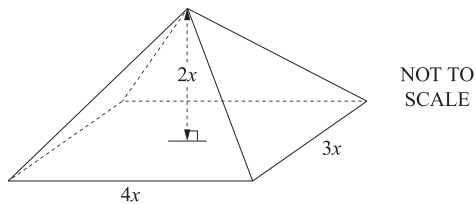


What is the approximate volume of concrete needed to make the water pipe?

- A. 0.06 m^3
- B. 0.09 m^3
- C. 0.70 m^3
- D. 0.99 m^3

16. Measurement, STD2 M1 2018 HSC 13 MC

A rectangular pyramid has base side lengths $3x$ and $4x$. The perpendicular height of the pyramid is $2x$. All measurements are in metres.

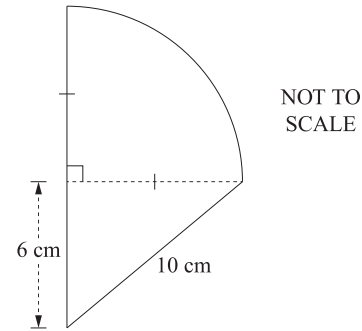


What is the volume of the pyramid in cubic metres?

- A. $8x^3$
- B. $9x^3$
- C. $12x^3$
- D. $24x^3$

17. Measurement, STD2 M1 2018 HSC 22 MC

A shape consisting of a quadrant and a right-angled triangle is shown.



What is the perimeter of this shape, correct to one decimal place?

- A. 28.6 cm
- B. 36.6 cm
- C. 66.3 cm
- D. 74.3 cm

18. Measurement, STD2 M1 SM-Bank 26 MC

During a flood, 12.5 hectares of land was covered by water to a depth of 30 cm.

How many kilolitres of water covered the land?

(1 hectare = 10 000 m² and 1 m³ = 1000 L)

- A. 3.75 kL
- B. 37.5 kL
- C. 37 500 kL
- D. 37 500 000 kL

19. Measurement, STD2 M1 2010 HSC 17 MC

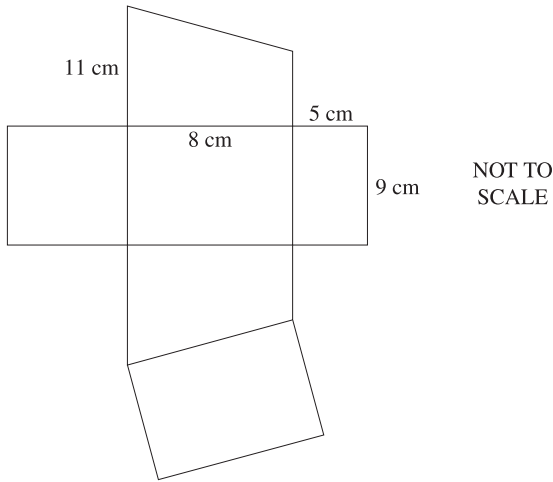
During a flood 1.5 hectares of land was covered by water to a depth of 17 cm.

How many kilolitres of water covered the land? (1 hectare = 10 000 m²)

- A. 2.55 kL
- B. 2550 kL
- C. 255 000 kL
- D. 2 550 000 kL

20. Measurement, STD2 M1 2013 HSC 25 MC

A net is made using four rectangles and two trapeziums. It is folded to form a solid.



What is the volume of the solid, in cm^3 ?

- A. 360 cm^3
- B. 434 cm^3
- C. 440 cm^3
- D. 576 cm^3

21. Measurement, STD2 M1 2008 HSC 21 MC

A sphere and a closed cylinder have the same radius.

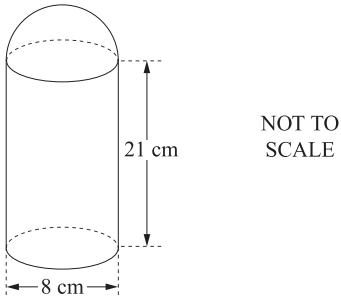
The height of the cylinder is four times the radius.

What is the ratio of the volume of the cylinder to the volume of the sphere?

- A. 2 : 1
- B. 3 : 1
- C. 4 : 1
- D. 8 : 1

22. Measurement, STD2 M1 2012 HSC 25 MC

The solid shown is made of a cylinder with a hemisphere (half a sphere) on top.



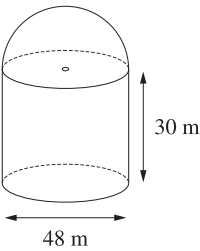
What is the total surface area of the solid, to the nearest square centimetre?

- A. 628 cm^2
- B. 679 cm^2
- C. 729 cm^2
- D. 829 cm^2

23. Measurement, STD2 M1 2014 HSC 25 MC

A *RAP Data (vs state mean) - Bottom 3%: School result (35%) was 5% below state average (40%)*

A grain silo is made up of a cylinder with a hemisphere (half a sphere) on top. The outside of the silo is to be painted.

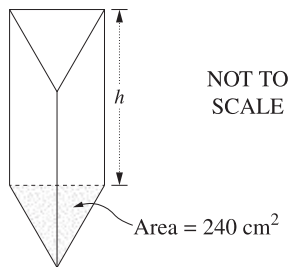


What is the area to be painted?

- A. 8143 m^2
- B. $11\,762 \text{ m}^2$
- C. $12\,667 \text{ m}^2$
- D. $23\,524 \text{ m}^2$

24. Measurement, STD2 M1 2016 HSC 12 MC

A container is in the shape of a triangular prism which has a capacity of 12 litres. The area of the base is 240 cm^2 .

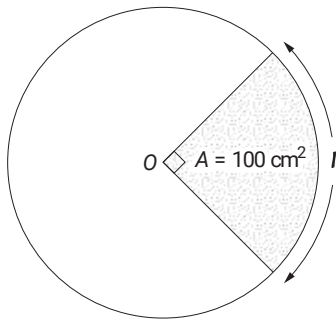


What is the distance, h , between the two triangular ends of the container?

- A. 5 cm
- B. 20 cm
- C. 25 cm
- D. 50 cm

25. Measurement, STD2 M1 2017 HSC 25 MC

In the circle, centre O , the area of the quadrant is 100 cm^2 .



What is the arc length l , correct to one decimal place?

- A. 8.9 cm
- B. 11.3 cm
- C. 17.7 cm
- D. 25.1 cm

26. Measurement, STD2 M1 2018 HSC 24 MC

The coordinates of city A are $(39^\circ\text{N}, 75^\circ\text{W})$. City B lies on the same longitude and is 5700 km south of city A .

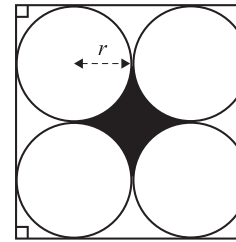
What is the latitude of city B , given the earth's radius is 6400 km?

- A. 51°N
- B. 51°S
- C. 12°N
- D. 12°S

27. Measurement, STD2 M1 SM-Bank 2 MC

Four identical circles of radius r are drawn inside a square, as shown in the diagram below

The region enclosed by the circles has been shaded in the diagram.

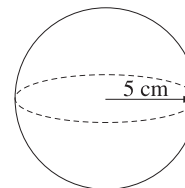


The shaded area can be found using

- A. $4r^2 - 2\pi r$
- B. $4r^2 - \pi r^2$
- C. $4r - \pi r^2$
- D. $2r^2 - \pi r^2$

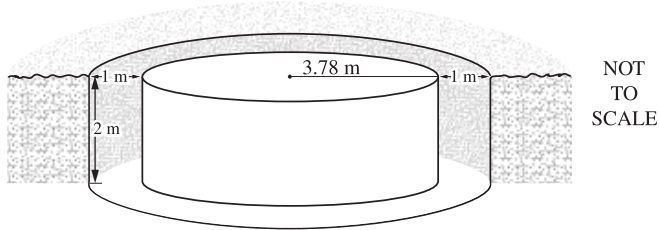
28. Measurement, STD2 M1 2016 HSC 26a

Calculate the surface area of a sphere with a radius of 5 cm, correct to the nearest whole number. (1 mark)



29. Measurement, STD2 M1 2007 HSC 23b

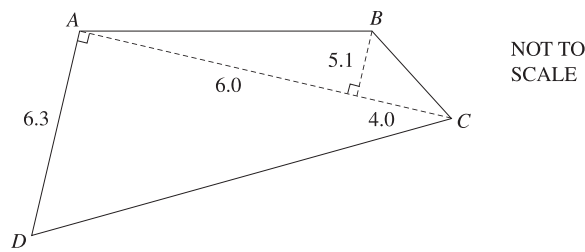
A cylindrical water tank, of height 2 m, is placed in the ground at a school.
The radius of the tank is 3.78 metres. The hole is 2 metres deep. When the tank is placed in the hole there is a gap of 1 metre all the way around the side of the tank.



- i. When digging the hole for the water tank, what volume of soil was removed? Give your answer to the nearest cubic metre. (3 marks)
- ii. Sprinklers are used to water the school oval at a rate of 7500 litres per hour.
The water tank holds 90 000 litres when full.
For how many hours can the sprinklers be used before a full tank is emptied? (1 mark)
- iii. Water is to be collected in the tank from the roof of the school hall, which has an area of 400 m².
During a storm, 20 mm of rain falls on the roof and is collected in the tank.
How many litres of water were collected? (2 marks)

30. Measurement, STD2 M1 2004 HSC 23a

The diagram shows the shape of Carmel’s garden bed. All measurements are in metres.



- i. Show that the area of the garden bed is 57 square metres. (2 marks)
- ii. Carmel decides to add a 5 cm layer of straw to the garden bed.
Calculate the volume of straw required. Give your answer in cubic metres. (2 marks)
- iii. Each bag holds 0.25 cubic metres of straw.
How many bags does she need to buy? (2 marks)
- iv. A straight fence is to be constructed joining point A to point B.
Find the length of this fence to the nearest metre. (2 marks)

31. Measurement, STD2 M1 SM-Bank 8

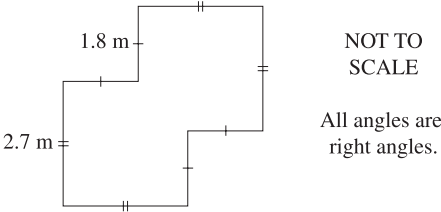


- A cannon ball is made out of steel and has a diameter of 23 cm.
- i. Find the volume of the sphere in cubic centimetres (correct to 1 decimal place). (2 marks)
 - ii. It is known that the mass of the steel used is 8.2 tonnes/m³. Use this information to find the mass of the cannon ball to the nearest gram. (2 marks)

32. Measurement, STD2 M1 2009 HSC 23c

⚡ Part ii: RAP Data (vs state mean) - Bottom 23%: School result (58%) was 5% above state average (53%)

The diagram shows the shape and dimensions of a terrace which is to be tiled.



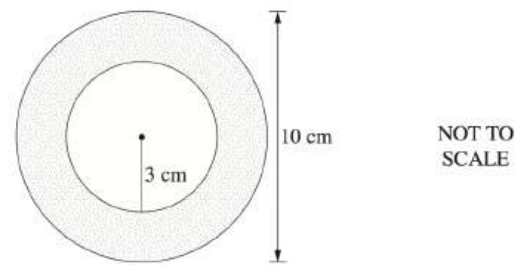
- i. Find the area of the terrace. (2 marks)
- ii. Tiles are sold in boxes. Each box holds one square metre of tiles and costs \$55. When buying the tiles, 10% more tiles are needed, due to cutting and wastage.
Find the total cost of the boxes of tiles required for the terrace. (2 marks)

33. Measurement, STD2 M1 2015 HSC 26f

- Approximately 71% of Earth’s surface is covered by water. Assume Earth is a sphere with a radius of 6400 km.
- Calculate the number of square kilometres covered by water. (2 marks)

34. Measurement, STD2 M1 2015 HSC 28a

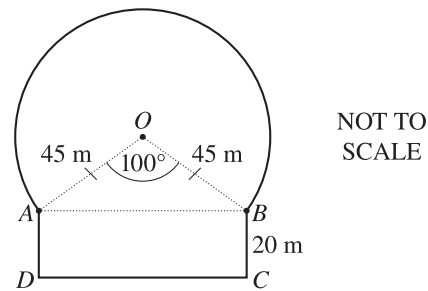
The diagram shows an annulus.



Calculate the area of the annulus. (1 mark)

35. Measurement, STD2 M1 2016 HSC 30c

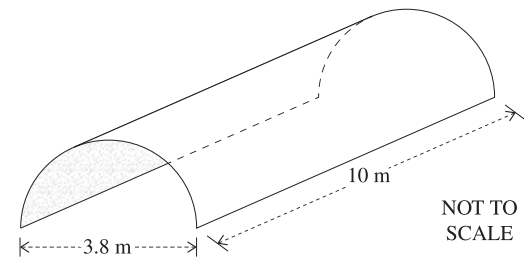
A school playground consists of part of a circle, with centre O , and a rectangle as shown in the diagram. The radius OB of the circle is 45 m, the width BC of the rectangle is 20 m and AOB is 100° .



What is the area of the whole playground, correct to the nearest square metre? (5 marks)

36. Measurement, STD2 M1 2018 HSC 27c

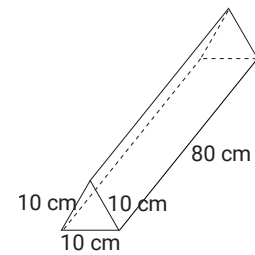
A shade shelter is to be constructed in the shape of half a cylinder with open ends. The diameter is 3.8 m and the length is 10 m.



The curved roof is to be made of plastic sheeting.
What area of plastic sheeting is required, to the nearest m^2 ? (2 marks)

37. Measurement, STD2 M1 SM-Bank 4

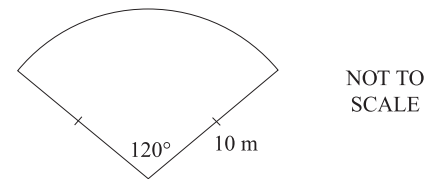
Steel rods are manufactured in the shape of equilateral triangular prisms.



- i. Find the volume of the prism (answer correct to 1 decimal place). (2 marks)
- ii. The mass of steel is 7850 kg/m^3 . Use this information to find the mass of the steel rod correct to the nearest gram. (2 marks)

38. Measurement, STD1 M1 2019 HSC 25

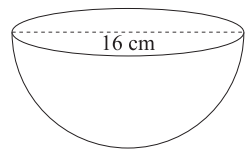
The diagram shows a sector with an angle of 120° cut from a circle with radius 10 m.



What is the perimeter of the sector? Write your answer correct to 1 decimal place. (3 marks)

39. Measurement, STD2 M1 2019 HSC 16

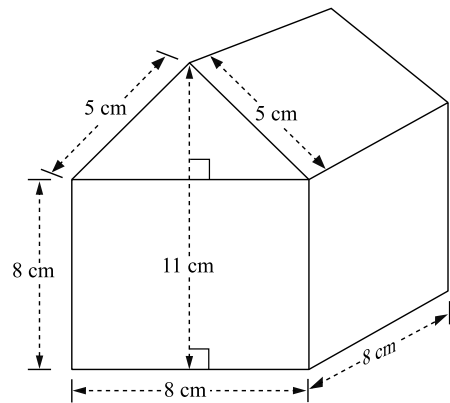
A bowl is in the shape of a hemisphere with a diameter of 16 cm.



What is the volume of the bowl, correct to the nearest cubic centimetre? (2 marks)

40. Measurement, STD2 M1 2020 HSC 25

A composite solid consists of a triangular prism which fits exactly on top of a cube, as shown.

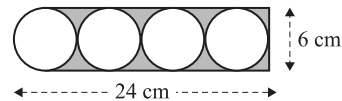


NOT TO SCALE

Find the surface area of the composite solid. (3 marks)

41. Measurement, STD2 M1 2021 FUR1 6

A child's toy has the following design.



Find the area of the shaded region to the nearest square centimetre. (2 marks)

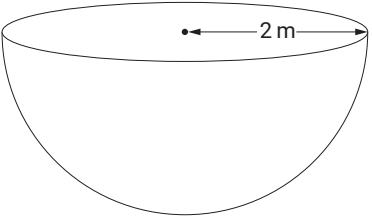
42. Measurement, STD2 M1 2021 HSC 16

The volume, V , of a sphere is given by the formula

$$V = \frac{4}{3}\pi r^3,$$

where r is the radius of the sphere.

A tank consists of the bottom half of a sphere of radius 2 metres, as shown.

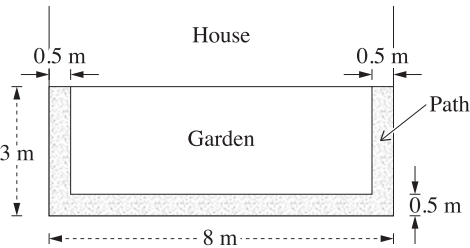


Find the volume of the tank in cubic metres, correct to one decimal place. (2 marks)

43. Measurement, STD2 M7 2023 HSC 26

⚡ Part b: RAP Data (vs state mean) - Bottom 16%: School result (64%) was 3% above state average (61%)

Kim is building a path around a garden at the back of a house, as shown. The path is 0.5 m wide.



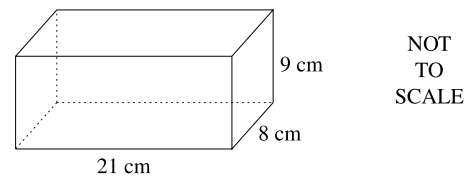
- a. Find the area of the path. (2 marks)
- b. Kim is mixing some concrete for the path. The concrete mix is made up of crushed rock, sand and cement in the ratio of 4 : 2 : 1 by weight.

Kim needs 2.1 tonnes of concrete in the correct ratio.

Calculate how many 15 kg bags of cement Kim needs to buy. (3 marks)

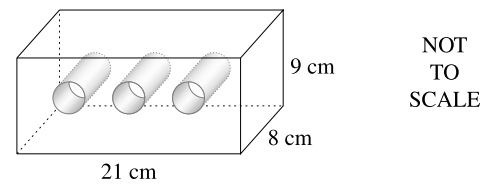
44. Measurement, STD2 M1 2005 HSC 23b

A clay brick is made in the shape of a rectangular prism with dimensions as shown.



i. Calculate the volume of the clay brick. (1 mark)

Three identical cylindrical holes are made through the brick as shown. Each hole has a radius of 1.4 cm.

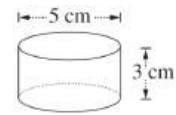


ii. What is the volume of clay remaining in the brick after the holes have been made? (Give your answer to the nearest cubic centimetre.) (3 marks)

iii. What percentage of clay is removed by making the holes through the brick? (Give your answer correct to one decimal place.) (1 mark)

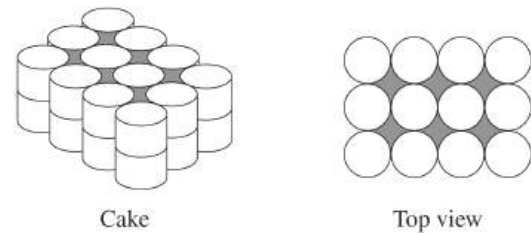
45. Measurement, STD2 M1 2016 HSC 28e

A company makes large marshmallows. They are in the shape of a cylinder with diameter 5 cm and height 3 cm, as shown in the diagram.



i. Find the volume of one of these large marshmallows, correct to one decimal place. (2 marks)

A cake is to be made by stacking 24 of these large marshmallows and filling the gaps between them with chocolate. The diagrams show the cake and its top view. The shading shows the gaps to be filled with chocolate.

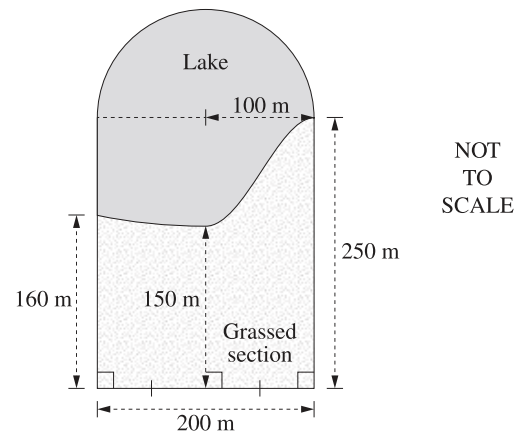


ii. What volume of chocolate will be required? Give your answer correct to the nearest whole number. (3 marks)

46. Measurement, STD2 M1 2022 HSC 32

The diagram shows a park consisting of a rectangle and a semicircle. The semicircle has a radius of 100 m. The dimensions of the rectangle are 200 m and 250 m.

A lake occupies a section of the park as shown. The rest of the park is a grassed section. Some measurements from the end of the grassed section to the edge of the lake are also shown.



a. Using two applications of the trapezoidal rule, calculate the approximate area of the grassed section. (2 marks)

b. Hence calculate the approximate area of the lake, to the nearest square metre. (2 marks)

47. Measurement, STD2 M1 2016 HSC 30a

The area of a roof is 30 m². Any rain that falls on the roof flows directly onto a garden.

Calculate how many litres of water flow onto the garden when 20 mm of rain falls on the roof. (2 marks)

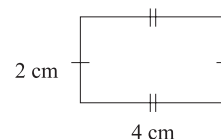
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Worked Solutions

1. Measurement, STD2 M1 2011 HSC 1 MC

$\Rightarrow D$

2. Measurement, STD2 M1 2019 HSC 1 MC



$$\text{Perimeter} = 2 \times (4 + 2) = 12 \text{ cm}$$

$\Rightarrow A$

3. Measurement, STD2 M1 2021 HSC 1 MC

Consider each option:

Option A: $4 \times 8 = 32 \text{ cm}$

Option B: $2 \times (3 + 11) = 28 \text{ cm}$

Option C: $3 \times 10 = 30 \text{ cm}$

Option D: $4 \times 2 + 3 + 9 = 20 \text{ cm}$

$\Rightarrow A$

4. Measurement, STD2 M1 2012 HSC 6 MC

Convert all measurements to centimetres:

$$V = l \times b \times h$$

$$= 30 \times 5 \times 400$$

$$= 60\,000 \text{ cm}^3$$

$\Rightarrow C$

5. Measurement, STD2 M1 2009 HSC 11 MC

Area = Area of Sector – Area of triangle

$$= \left(\frac{\theta}{360} \times \pi r^2 \right) - \left(\frac{1}{2} \times bh \right)$$

$$= \left(\frac{90}{360} \times \pi \times 8^2 \right) - \left(\frac{1}{2} \times 4 \times 4 \right)$$

$$= 50.2654 \dots - 8$$

$$= 42.265 \dots \text{ cm}^2$$

$\Rightarrow B$

6. Measurement, STD2 M1 2009 HSC 19 MC

Since diameter sphere = 12 cm
⇒ Radius of cylinder = 6 cm

Height of cylinder = $2 \times \text{diameter of sphere}$
= 2×12
= 24 cm

∴ Volume cylinder = $\pi r^2 h$
= $\pi \times 6^2 \times 24$
= 2714.336... cm³

⇒ B

7. Measurement, STD2 M1 2013 HSC 12 MC

Volume = Vol (cube) + Vol (pyramid)
= $l^3 + \frac{1}{3}Ah$
= $(9 \times 9 \times 9) + \left(\frac{1}{3} \times 9 \times 9 \times 10\right)$
= 999 cm³

⇒ B

8. Measurement, STD2 M1 2013 HSC 16 MC

Shaded area = Area of segment – Area of rectangle
= $\frac{1}{4}\pi r^2 - (6 \times 2)$
= $\frac{1}{4}\pi \times 9^2 - 12$
= 51.617... cm²

⇒ B

9. Measurement, STD2 M1 2013 HSC 19 MC

Area = $\frac{1}{2} \times \pi(R^2 - r^2)$
= $\frac{1}{2} \times \pi(5^2 - 2^2)$
= $\frac{21}{2}\pi$
= 33 cm² (nearest cm²)

⇒ B

10. Measurement, STD2 M1 2008 HSC 11 MC

Area = Square – Circle
= $(100 \times 100) - (\pi \times 5^2)$
= 10 000 – 78.5398...
= 9921.46... cm²

COMMENT: Students should see that answers are all in *cm*², and therefore use *cm* as the base unit for their calculations.

⇒ B

11. Measurement, STD2 M1 2008 HSC 2 MC

Surface Area
= $2 \times (3 \times 5) + 2 \times (2 \times 3) + (5 \times 2)$
= 30 + 12 + 10
= 52 cm²

⇒ C

12. Measurement, STD2 M1 2014 HSC 12 MC

Area of annulus
= $\pi(R^2 - r^2)$
= $\pi(4.5^2 - 3^2)$
= $\pi(11.25)$
= 35.3 m² (1 d.p.)

⇒ C

13. Measurement, STD2 M1 2015 HSC 8 MC

$$V = \frac{1}{3}Ah$$

$$\begin{aligned} A &= 35 \times 35 \\ &= 1225 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \therefore V &= \frac{1}{3} \times 1225 \times 22 \\ &= 8983.33 \dots \text{ m}^3 \end{aligned}$$

$$\Rightarrow C$$

14. Measurement, STD2 M1 2017 HSC 18 MC

$$\begin{aligned} \text{Area of trapezoid} &= \frac{1}{2}h(a + b) \\ &= \frac{1}{2} \times 1.2 \times (3.6 + 2.4) \\ &= 3.6 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \therefore \text{Volume} &= Ah \\ &= 3.6 \times 1.5 \\ &= 5.4 \text{ m}^3 \end{aligned}$$

$$\Rightarrow A$$

15. Measurement, STD2 M1 2017 HSC 22 MC

$$\begin{aligned} \text{Volume} &= \text{Area of annulus} \times h \\ &= (\pi R^2 - \pi r^2) \times 2.8 \\ &= (\pi \times 0.45^2 - \pi \times 0.35^2) \times 2.8 \\ &= 0.7037 \dots \\ &= 0.70 \text{ m}^3 \end{aligned}$$

$$\Rightarrow C$$

16. Measurement, STD2 M1 2018 HSC 13 MC

$$\begin{aligned} \text{Volume} &= \frac{1}{3}Ah \\ &= \frac{1}{3}(4x \times 3x \times 2x) \\ &= 8x^3 \end{aligned}$$

$$\Rightarrow A$$

17. Measurement, STD2 M1 2018 HSC 22 MC

Using Pythagoras to find radius (r):

$$\begin{aligned} r &= \sqrt{10^2 - 6^2} \\ &= \sqrt{64} \\ &= 8 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{Arc length} &= \frac{1}{4} \times 2\pi r \\ &= \frac{1}{4} \times 2 \times \pi \times 8 \\ &= 12.56 \dots \text{ cm} \end{aligned}$$

$$\begin{aligned} \therefore \text{Perimeter} &= 8 + 6 + 10 + 12.56 \dots \\ &= 36.57 \dots \end{aligned}$$

$$\Rightarrow B$$

18. Measurement, STD2 M1 SM-Bank 26 MC

Volume of water in m³:

$$\begin{aligned} \text{Vol} &= Ah \\ &= 12.5 \times 10\,000 \times 30 \text{ cm} \\ &= 12.5 \times 10\,000 \times 0.3 \\ &= 37\,500 \text{ m}^3 \\ &= 37\,500 \text{ kL} \end{aligned}$$

$$\Rightarrow C$$

19. Measurement, STD2 M1 2010 HSC 17 MC

Area covered = $1.5 \times 10\,000$
= $15\,000\text{ m}^2$

Volume = Ah
= $15\,000 \times 0.17$
= 2550 m^3

$1\text{ m}^3 = 1000\text{ L} = 1\text{ kL}$
 $\therefore \text{Volume} = 2550\text{ kL}$
 $\Rightarrow B$

20. Measurement, STD2 M1 2013 HSC 25 MC

Volume = Ah where A is the area of a trapezium

$A = \frac{1}{2}h(a + b)$
 $= \frac{1}{2} \times 8(11 + 5)$
 $= 64\text{ cm}^2$

$\therefore V = Ah = 64 \times 9 = 576\text{ cm}^3$
 $\Rightarrow D$

21. Measurement, STD2 M1 2008 HSC 21 MC

$V_{\text{cylinder}} : V_{\text{sphere}}$
 $\pi r^2 h : \frac{4}{3}\pi r^3$
 $\underbrace{\pi r^2 4r}_{h=4r} : \frac{4}{3}\pi r^3$
 $4\pi r^3 : \frac{4}{3}\pi r^3$
 $3 : 1$

$\Rightarrow B$

♦♦ Mean mark 34%
NOTE: The unit conversion $1\text{ m}^3 = 1000\text{ L}$ is contained in the Formulae and Data sheet given out in the exam.

♦ Mean mark 35%
COMMENT: Note that h in the "area" formula is different to the h used in the "volume" formula.

♦♦ Mean mark 33%.

22. Measurement, STD2 M1 2012 HSC 25 MC

Total surface area
 $= \pi r^2 + 2\pi rh + \frac{1}{2} \times 4\pi r^2$
 $= \pi \times 4^2 + 2\pi \times 4 \times 21 + \frac{1}{2} \times 4\pi \times 4^2$
 $= 678.58\dots\text{ cm}^2$
 $\Rightarrow B$

23. Measurement, STD2 M1 2014 HSC 25 MC

Total Area = Area of cylinder + $\frac{1}{2}$ sphere

Area of cylinder = $2\pi rh$
 $= 2\pi \times 24 \times 30$
 $= 4523.9$

Area of $\frac{1}{2}$ sphere = $\frac{1}{2} \times 4\pi r^2$
 $= \frac{1}{2} \times 4\pi \times 24^2$
 $= 3619.1$

$\therefore \text{Total area} = 4523.9 + 3619.1$
 $= 8143\text{ m}^2$

$\Rightarrow A$

24. Measurement, STD2 M1 2016 HSC 12 MC

$1\text{ mL} = 1\text{ cm}^3 \Rightarrow 1\text{ L} = 1000\text{ cm}^3$

Volume = Ah
 $12\,000 = 240 \times h$
 $h = \frac{12\,000}{240}$
 $= 50\text{ cm}$

$\Rightarrow D$

♦ Mean mark 40%

♦♦ Mean mark 35%.

25. Measurement, STD2 M1 2017 HSC 25 MC

Find r :

$$\text{Area} = \frac{1}{4}\pi r^2$$

$$100 = \frac{1}{4}\pi r^2$$

$$r^2 = \frac{400}{\pi}$$

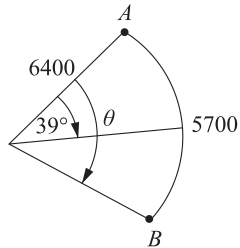
$$\therefore r = 11.283\dots \text{ cm}$$

♦ Mean mark 44%.

$$\begin{aligned}\text{Arc length} &= \frac{\theta}{360} \times 2\pi r \\ &= \frac{90}{360} \times 2\pi \times 11.283\dots \\ &= 17.724\dots \\ &= 17.7 \text{ cm}\end{aligned}$$

$\Rightarrow C$

26. Measurement, STD2 M1 2018 HSC 24 MC



$$\text{Circumference} = 2 \times \pi \times 6400$$

$$\frac{\theta}{360} = \frac{5700}{2\pi \times 6400}$$

$$\theta = \frac{5700 \times 360}{2\pi \times 6400}$$

$$\approx 51^\circ$$

$\therefore$ City B is 12° South.

$\Rightarrow D$

♦ Mean mark 45%.

COMMENT: If the earth's radius is stated, this past question is arguably within the scope of the new syllabus.

27. Measurement, STD2 M1 SM-Bank 2 MC

$$\text{Area of square} = 4r \times 4r = 16r^2$$

$$\text{Area of circles} = 4\pi r^2$$

Shaded Area

$$= \frac{1}{4} \times (16r^2 - 4\pi r^2)$$

$$= 4r^2 - \pi r^2$$

$\Rightarrow B$

28. Measurement, STD2 M1 2016 HSC 26a

$$SA = 4\pi r^2$$

$$= 4 \times \pi \times 5^2$$

$$= 314.15\dots$$

$$= 314 \text{ cm}^2 \text{ (nearest cm}^2\text{)}$$

29. Measurement, STD2 M1 2007 HSC 23b

i. $V = \pi r^2 h$ where

$$h = 2 \text{ and } r = 4.78 \text{ m}$$

$$\therefore V = \pi \times 4.78^2 \times 2$$

$$= 143.56\dots$$

$$= 144 \text{ m}^3 \text{ (nearest m}^3\text{)}$$

ii. Total water = 90 000 litres

$$\text{Usage} = 7500 \text{ litres/hr}$$

$\therefore$ Hours before it is empty

$$= \frac{90\,000}{7500}$$

$$= 12 \text{ hours}$$

iii. Water collected

$$= 400 \times 0.020$$

$$= 8 \text{ m}^2$$

$$= 8000 \text{ litres}$$

30. Measurement, STD2 M1 2004 HSC 23a

i. Area of $\triangle ABC = \frac{1}{2} \times b \times h$

$$= \frac{1}{2} \times 10 \times 5.1$$
$$= 25.5 \text{ m}^2$$

Area of $\triangle ACD = \frac{1}{2} \times 10 \times 6.3$

$$= 31.5 \text{ m}^2$$

$\therefore$ Total Area = 25.5 + 31.5

$$= 57 \text{ m}^2 \dots \text{as required}$$

ii. $V = Ah$

$$= 57 \times 0.05$$
$$= 2.85 \text{ m}^3$$

iii. Bags to buy = $\frac{2.85}{0.25}$

$$= 11.4$$

$\therefore$ She needs to buy 12 bags.

iv. Using Pythagoras,

$$AB^2 = 6.0^2 + 5.1^2$$
$$= 36 + 26.01$$
$$= 62.01$$
$$AB = 7.874\dots$$
$$= 8 \text{ m (nearest metre)}$$

31. Measurement, STD2 M1 SM-Bank 8

i. Radius = $\frac{23}{2} = 11.5 \text{ cm}$

Volume = $\frac{4}{3}\pi r^3$

$$= \frac{4}{3} \times \pi \times 11.5^3$$
$$= 6370.626\dots$$
$$= 6370.6 \text{ cm}^3 \text{ (to 1 d.p.)}$$

ii. Convert m^3 to cm^3 :

$$1 \text{ m}^3 = 100 \text{ cm} \times 100 \text{ cm} \times 100 \text{ cm}$$
$$= 1\,000\,000 \text{ cm}^3$$

Convert 8.2 tonnes to grams:

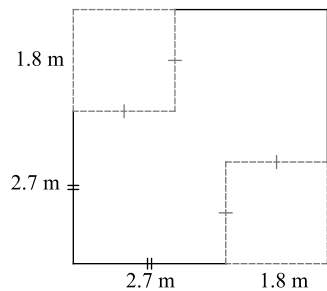
$$8.2 \text{ tonnes} = 8200 \text{ kg}$$
$$= 8\,200\,000 \text{ g}$$

$\therefore$ Weight of cannon ball

$$= 6370.6 \times \frac{8\,200\,000}{1\,000\,000}$$
$$= 52\,238.92$$
$$= 52\,239 \text{ grams}$$

32. Measurement, STD2 M1 2009 HSC 23c

i.



$$\begin{aligned}
 \text{Area} &= \text{Area of big square} - \text{Area of 2 cut-out squares} \\
 &= (2.7 + 1.8) \times (2.7 + 1.8) - 2 \times (1.8 \times 1.8) \\
 &= 20.25 - 6.48 \\
 &= 13.77 \text{ m}^2
 \end{aligned}$$

ii. Tiles required = $(13.77 + 10\%) \times 13.77$
 $= 15.147 \text{ m}^2$

$\Rightarrow$ 16 boxes are needed

$$\begin{aligned}
 \therefore \text{Total cost of boxes} &= 16 \times \$55 \\
 &= \$880
 \end{aligned}$$

33. Measurement, STD2 M1 2015 HSC 26f

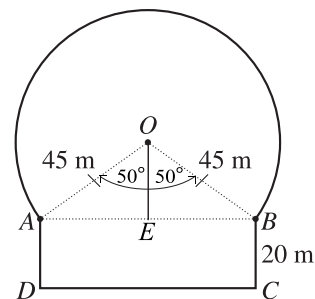
$$\begin{aligned}
 \text{Surface area of Earth} \\
 &= 4\pi r^2 \\
 &= 4\pi \times 6400^2
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Surface covered by water} \\
 &= 71\% \times 4\pi \times 6400^2 \\
 &= 365\,450\,163.7\dots \\
 &= 3.7 \times 10^8 \text{ km}^2 \text{ (nearest km}^2\text{)}
 \end{aligned}$$

34. Measurement, STD2 M1 2015 HSC 28a

$$\begin{aligned}
 \text{Area of annulus} \\
 &= \pi(R^2 - r^2) \\
 &= \pi(5^2 - 3^2) \\
 &= \pi(25 - 9) \\
 &= 16\pi \text{ cm}^2 \\
 &\approx 50.26\dots \text{ cm}^2
 \end{aligned}$$

35. Measurement, STD2 M1 2016 HSC 30c



In $\triangle OEB$,

$$\begin{aligned}
 \sin 50^\circ &= \frac{EB}{45} \\
 EB &= 45 \times \sin 50^\circ \\
 &= 34.47\dots \\
 \therefore AB &= 2 \times 34.47\dots \\
 &= 68.944 \text{ (3 d.p.)}
 \end{aligned}$$

$$\begin{aligned}
 \cos 50^\circ &= \frac{OE}{45} \\
 \therefore OE &= 45 \times \cos 50^\circ \\
 &= 28.925 \text{ (3 d.p.)}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of } \triangle OAB \\
 &= \frac{1}{2} \times AB \times OE \\
 &= \frac{1}{2} \times 68.944 \times 28.925 \\
 &= 997.12 \text{ m}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Area } ABCD &= 20 \times 68.944 \\
 &= 1378.88 \text{ m}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of major sector } OAB \\
 &= \pi \times 45^2 \times \frac{260}{360} \\
 &= 4594.58 \text{ m}^2
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{Area of playground} \\
 &= 997.12 + 1378.88 + 4594.58 \\
 &= 6970.58 \\
 &= 6971 \text{ m}^2 \text{ (nearest m}^2\text{)}
 \end{aligned}$$

36. Measurement, STD2 M1 2018 HSC 27c

Flatten out the half cylinder,

$$\begin{aligned}\text{Width} &= \frac{1}{2} \times \text{circumference} \\ &= \frac{1}{2} \times \pi \times 3.8 \\ &= 5.969\dots\end{aligned}$$

$$\begin{aligned}\therefore \text{Sheeting required} &= 10 \times 5.969\dots \\ &= 59.69\dots \\ &= 60 \text{ m}^2 \text{ (nearest m}^2\text{)}\end{aligned}$$

37. Measurement, STD2 M1 SM-Bank 4

i. Area of triangular face (using sine rule)

$$\begin{aligned}&= \frac{1}{2} \times 10 \times 10 \times \sin 60^\circ \\ &= 43.301\dots\end{aligned}$$

$$\begin{aligned}\text{Volume} &= Ah \\ &= 43.301\dots \times 80 \\ &= 3464.10\dots \\ &= 3464.1 \text{ cm}^3 \text{ (to 1 d.p.)}\end{aligned}$$

ii. Converting kg to g:

$$7850 \text{ kg} = 7\,850\,000 \text{ g}$$

Converting m³ to cm³:

$$\begin{aligned}1 \text{ m}^3 &= 100 \text{ cm} \times 100 \text{ cm} \times 100 \text{ cm} \\ &= 1\,000\,000 \text{ cm}^3\end{aligned}$$

$$\begin{aligned}\therefore \text{Weight of steel rod} &= 3464.1 \times \frac{7\,850\,000}{1\,000\,000} \\ &= 27\,193.185 \\ &= 27\,193 \text{ g (nearest gram)}\end{aligned}$$

38. Measurement, STD1 M1 2019 HSC 25

$$\begin{aligned}\text{Arc length} &= \frac{120}{360} \times 2 \times \pi \times 10 \\ &= 20.94\end{aligned}$$

$$\begin{aligned}\therefore \text{Perimeter} &= 20.94 + 2 \times 10 \\ &= 40.94 \\ &= 40.9 \text{ (1 d. p.)}\end{aligned}$$

39. Measurement, STD2 M1 2019 HSC 16

$$\begin{aligned}V &= \frac{1}{2} \times \frac{4}{3} \pi r^3 \\ &= \frac{1}{2} \times \frac{4}{3} \times \pi \times 8^3 \\ &= 1072.3\dots \\ &= 1072 \text{ cm}^3 \text{ (nearest cm}^3\text{)}\end{aligned}$$

40. Measurement, STD2 M1 2020 HSC 25

$$\begin{aligned}\text{S.A. of 1 face of cube} &= 8 \times 8 = 64 \text{ cm}^2 \\ \text{Height of triangle} &= 11 - 8 = 3 \text{ cm}\end{aligned}$$

$$\begin{aligned}\therefore \text{S.A. (triangular prism)} &= 2 \times \left(\frac{1}{2} \times 8 \times 3 \right) + 2 \times (5 \times 8) \\ &= 24 + 80 \\ &= 104 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\therefore \text{Total S.A.} &= 5 \times 64 + 104 \\ &= 424 \text{ cm}^2\end{aligned}$$

41. Measurement, STD2 M1 2021 FUR1 6

Circle radius = 3 cm
Consider the rectangle starting from the middle of the left circle.

$$\begin{aligned}\text{Shaded Area} &= \text{Area rectangle} - 3.5 \times \text{Area circle} \\ &= (21 \times 6) - 3.5 \times \pi \times 3^2 \\ &= 27.03\dots \text{ cm}^2 \\ &= 27 \text{ cm}^2 \text{ (nearest whole)}\end{aligned}$$

42. Measurement, STD2 M1 2021 HSC 16

$$\begin{aligned} V &= \frac{1}{2} \times \frac{4}{3} \pi r^3 \\ &= \frac{1}{2} \times \frac{4}{3} \times \pi \times 2^3 \\ &= 16.755\dots \\ &= 16.8 \text{ m}^3 \text{ (1 d.p.)} \end{aligned}$$

43. Measurement, STD2 M7 2023 HSC 26

- a.** Area outer rectangle $= 3 \times 8 = 24 \text{ m}^2$
Area garden $= 2.5 \times 7 = 17.5 \text{ m}^2$
 $A_{\text{path}} = 24 - 17.5$
 $= 6.5 \text{ m}^2$
- b.** 7 parts $= 2.1$ tonnes
 $1 \text{ part} = \frac{2.1}{7} = 0.3 \text{ tonnes} = 300 \text{ kgs}$
Rock : Sand : Cement $= 4 : 2 : 1 = 1200 : 600 : 300$
 $\Rightarrow$ 300 kgs of cement are required
 $\therefore$ Bags of cement $= \frac{300}{15}$
 $= 20 \text{ bags}$

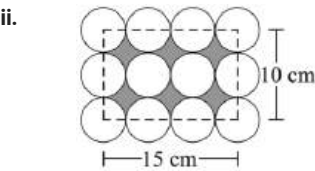
44. Measurement, STD2 M1 2005 HSC 23b

- i.** $V = l \times b \times h$
 $= 21 \times 8 \times 9$
 $= 1512 \text{ cm}^3$
- ii.** Volume of each hole
 $= \pi r^2 h$
 $= \pi \times 1.4^2 \times 8$
 $= 49.260\dots \text{ cm}^3$

 $\therefore$ Volume of clay still in brick
 $= 1512 - (3 \times 49.260\dots)$
 $= 1364.219\dots$
 $= 1364 \text{ cm}^3 \text{ (nearest whole)}$
- iii.** Percentage of clay removed
 $= \frac{(3 \times 49.260\dots)}{1512} \times 100$
 $= 9.773\dots$
 $= 9.8\% \text{ (1 d.p.)}$

45. Measurement, STD2 M1 2016 HSC 28e

i. $V = \pi r^2 h$
 $= \pi \times 2.5^2 \times 3$
 $= 58.904\dots$
 $= 58.9 \text{ cm}^3 \text{ (1 d.p.)}$



Volume of rectangle
 $= 15 \times 10 \times 6$
 $= 900 \text{ cm}^3$

Volume of marshmallows in rectangle
 $= 6 \times 2 \times 58.9$
 $= 706.8 \text{ cm}^3$

$\therefore$ Volume of chocolate
 $= 900 - 706.8$
 $= 193.2$
 $= 193 \text{ cm}^3 \text{ (nearest cm}^3\text{)}$

♦ Mean mark part (ii) 35%.

46. Measurement, STD2 M1 2022 HSC 32

a. $h = 100 \text{ m}$
 $A \approx \frac{h}{2}(x_1 + x_2) + \frac{h}{2}(x_2 + x_3)$
 $\approx \frac{100}{2}(160 + 150) + \frac{100}{2}(150 + 250)$
 $\approx 50 \times 310 + 50 \times 400$
 $\approx 35\,500 \text{ m}^2$

b. Total Area = Area of Rectangle + Area of Semi-Circle
 $= (250 \times 200) + \frac{1}{2} \times \pi \times 100^2$
 $= 65\,707.96 \text{ m}^2$

♦ Mean mark (b) 44%.

Area of Lake = $67\,708 - 35\,500$
 $= 30\,208 \text{ m}^2$

47. Measurement, STD2 M1 2016 HSC 30a

Volume = Ah
 $= 30 \times \frac{20}{1000}$
 $= 0.6 \text{ m}^3$
 $= 600 \text{ L}$

♦♦ Mean Mark 27%.
STRATEGY: Converting 20 mm into metres as a 1st step is an efficient approach here.